

Investigation into phenomena surrounding universally Baire sets

Soutenance de thèse

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- 1 Introduction
- 2 Local Definability of HOD in $L(\mathbb{R})$
- 3 Constructing Universally Baire Sets
- 4 Generic Absoluteness and Maximality
- 5 SSP-satisfiability

1 Introduction

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Global Assumption

We are assuming, if not stated otherwise, that ZFC holds and there exists a proper class of Woodin cardinals.

Theorem (Woodin)

Let A be a universally Baire set. Then

- a. $(\mathbb{R}, A)^\#$ is universally Baire,*
- b. for every V -generic filter g , $((\mathbb{R}^V, A)^\#)_g = (\mathbb{R}_g, A_g)^\#$.*

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Corollary (Martin, Steel, Woodin)

- a. $L(\mathbb{R}) \models \text{AD}^+$
- b. *For every V -generic filter g , there is an elementary embedding $j : L(\mathbb{R}^V) \rightarrow L(\mathbb{R}_g)$.*

Question

- a. What more can be said about the theory of $L(\mathbb{R})$?
- b. Does model $L(\mathbb{R})$ relate naturally to some model of ZFC?

Theorem (Gödel, Vopěnka)

Assume ZF holds. Then

- a. $\text{HOD} \models \text{ZFC}$,*
- b. for all $x \subseteq \text{Ord}$, there exists a HOD-generic filter g such that $x \in \text{HOD}[g] = \text{HOD}_g$.*

Theorem (Woodin)

Assume $\text{ZF} + \text{AD} + V = L(\mathbb{R})$. Then V is a symmetric extension of HOD .

Theorem (Steel)

$(\text{HOD} \parallel \Theta)^{L(\mathbb{R})}$ is a premouse.

Quote (Steel)

$\text{HOD}^{L(\mathbb{R})}$ is a core model below Θ .

Question

Is HOD locally definable below Θ in $L(\mathbb{R})$?

Theorem (Schindler)

Assume ZFC. Suppose

- 1. Ω is a measurable cardinal,*
- 2. for some $n < \omega$, V_Ω is closed for $\mathcal{M}_n^\#(-)$, but $\mathcal{M}_{n+1}^\#$ does not exist,*
- 3. K is the core model in V_Ω ,*
- 4. $\alpha \geq \aleph_2^V$ is a cardinal in K .*

Then $K \parallel \alpha^{+K}$ is definable (as a predicate) over $H(|\alpha|^+)$ without additional parameters.

Theorem (K.)

Suppose we are working in $L(\mathbb{R})$ and

1. $\eta \in [\delta_1^2, \Theta)$ is a cardinal in HOD,
2. μ_η is the (quasi-)club filter on $[\eta]^\omega$,
3. χ is the second ordinal such that $L((\text{HOD} \parallel \eta)^\omega)[\mu_\eta]$ satisfies $\text{ZF}^- + \text{“}\mathcal{P}([\eta]^\omega)\text{ exists”}$.

Then $\text{HOD} \parallel \eta^{+\text{HOD}}$ is definable (as a predicate) over

$$(L((\text{HOD} \parallel \eta)^\omega)[\mu_\eta], \in, \mu_\eta, (\text{HOD} \parallel \eta)^\omega),$$

without additional parameters.

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Definition

- a. Define A_0 to be $\mathbb{R}$.
- b. For $n < \omega$, define A_{n+1} to be $(\mathbb{R}, A_n)^\#$.

Corollary

For all $n < \omega$, A_n is universally Baire.

Question

What should A_ω be?

Definition

Let A_ω consist of all $n \frown x \in \mathbb{R}$, where $n < \omega$ and $x \in A_n$.

Proposition

A_ω is universally Baire and Wadge-above every A_n (for $n < \omega$).

Remark

If $\alpha < \omega_1$ is limit and $(A_\xi : \xi < \alpha)$ is defined, we can define A_α by

- picking a cofinal sequence $(\xi_n : n < \omega)$ in α ,
- joining $(A_{\xi_n} : n < \omega)$ as in the case ω .

Definition

The sequence $(A_\alpha : \alpha < \omega_1)$ is defined as in the remark.

Proposition

For all $\alpha < \omega_1$, A_α is universally Baire. Moreover, this sequence is strictly Wadge-increasing.

Question

What should A_{ω_1} be?

Remark

We cannot join the sequence $(A_\alpha : \alpha < \omega_1)$ in a way that will yield a universally Baire set because the sequence is not canonically defined.

Theorem (Woodin)

Let Σ be the strategy for $\mathcal{M}_\omega^\#$. Then $L(\mathbb{R}, \mathbb{R}^\#) = L(\mathbb{R}, \Sigma)$.

Theorem (Sargsyan)

There exists a universally Baire set above all A_α ($\alpha < \omega_1$) that codes a hod pair.

Remark

We can take A_{ω_1} to be a universally Baire set coding the mouse-least hod pair above all A_α ($\alpha < \omega_1$).

Definition

Let $(A_\alpha : \alpha < \mathfrak{F})$ be the longest sequence defined as above:

- a. at successor stages, we take the sharp;
- b. at limit stages of countable cofinality, we take the join of a cofinal ω -sequence;
- c. at limit stages of uncountable cofinality, we take a universally Baire set that codes a mouse-least hod pair which is Wadge-above the sets so far, if there is such a hod pair.

Theorem* (Sargsyan)

Either

a. there exists a universally Baire set that codes a hod pair with a Woodin cardinal that is a limit of Woodin cardinals

or

b. the sequence $(A_\alpha : \alpha < \mathfrak{F})$ is Wadge-cofinal among universally Baire sets.

Question

Is there a universally Baire set that codes a hod pair with a Woodin cardinal that is a limit of Woodin cardinals?

Theorem (K., Sargsyan)

Suppose there exists a proper class of Woodin cardinals that are limits of Woodin cardinals. Then there exists a universally Baire set that codes a hod pair with a measurable cardinal which is a limit of Woodin cardinals.

Corollary

Suppose there exists a proper class of Woodin cardinals that are limits of Woodin cardinals. Then there exists an inner model of $\text{ZFC} + V = \text{Ultimate-}L + \text{“there exists a proper class of Woodin cardinals”}$.

Corollary (of the proof)

Suppose there exists a proper class of Woodin cardinals that are limits of Woodin cardinals. Then the minimum model of $\text{AD}_{\mathbb{R}} + \text{“}\Theta \text{ is regular”}$ is contained in the Chang-plus model.

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Corollary

Suppose

- 1. A is a universally Baire set,*
- 2. ϕ is a sentence in the language $\{R_\in^2, R_A^1\}$,*
- 3. $\mathbb{P}$ is a poset.*

Then $V \models ((H_{\omega_1}, \in, A) \models \phi)$ if and only if $V^\mathbb{P} \models ((H_{\omega_1}, \in, A_G) \models \phi)$.

Theorem (K.)

Suppose

1. A is a universally Baire set,
2. ϕ is a Σ_1 sentence in the language $\{R_{\in}^2, R_{\text{NS}_{\omega_1}}^1, R_A^1\}$,
3. $\mathbb{P}$ is a poset.

Then $V \models ((H_{\omega_2}, \in, \text{NS}_{\omega_1}, A) \models \phi)$ if and only if $V^{\mathbb{P}} \models ((H_{\omega_2}, \in, \text{NS}_{\omega_1}, A_{\dot{G}}) \models \phi)$.

Remark

There cannot be Π_2 absoluteness of this form because $\neg\text{CH}$ is Π_2 over $(H_{\omega_2}, \in, \mathbb{R})$!

Definition

The sentence $(\dagger)_{\text{UB}}$ asserts that for every universally Baire set A and every Π_2 sentence ϕ in the language $\{R_{\in}^2, R_{\text{NS}_{\omega_1}}^1, R_A^1\}$, if

$$\exists \mathbb{P}, \Vdash_{\mathbb{P}} \left((H(\omega_2), \in, \text{NS}_{\omega_1}, A_{\dot{g}}) \models \phi \right),$$

then $(H(\omega_2), \in, \text{NS}_{\omega_1}, A) \models \phi$.

Definition

The sentence $(\dagger)_{\text{UB}}^-$ asserts that for every universally Baire set A and every Π_2 sentence ϕ in the language $\{R_{\in}^2, R_A^1\}$, if

$$\exists \mathbb{P}, \Vdash_{\mathbb{P}} \left((H(\omega_2), \in, A_{\dot{g}}) \models \phi \right),$$

then $(H(\omega_2), \in, A) \models \phi$.

Definition

The sentence **UB – BMM⁺⁺** asserts that for every universally Baire set A , every Σ_1 formula $\phi(v_a)$ in the language $\{R_\infty^2, R_{NS_{\omega_1}}^1, R_A^1\}$ and every $a \in H_{\omega_2}$, if

$$\exists \mathbb{P} \text{ ssp}, V^{\mathbb{P}} \models \left((H(\omega_2), \in, NS_{\omega_1}, A_{\dot{g}}) \models \phi[a] \right),$$

then $(H(\omega_2), \in, NS_{\omega_1}, A) \models \phi[a]$.

Definition

The sentence **UB – BMM** asserts that for every universally Baire set A , every Σ_1 formula $\phi(v_a)$ in the language $\{R_\infty^2, R_A^1\}$ and every $a \in H_{\omega_2}$, if

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then $(H(\omega_2), \in, A) \models \phi[a]$.

Theorem (Asperó, Schindler, Woodin)

$$\text{UB} - \text{BMM}^{++} \implies (\dagger)_{\text{UB}}$$

Theorem (K.)

$$\text{UB} - \text{BMM} \implies (\dagger)_{\text{UB}}^-$$

Remark

The proofs raise the question when a model to an infinitary propositional formula can be added by a stationary-set-preserving poset.

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Theorem (Väänänen?)

Let ϕ be an infinitary propositional in negation normal form. Then there exists a poset that forces a model for ϕ if and only if Player II has a winning strategy in the Model Existence Game.

Theorem (K., Veličković)

Let ϕ be an infinitary propositional in negation normal form. Then the following are equivalent.

- a. There exists an ssp poset that forces a model for ϕ .
- b. For some inaccessible κ , Player II has a winning strategy in $\mathfrak{D}_{\kappa, \phi}^{\text{ssp}}$ (a variation of the Model Existence Game).

Definition

Suppose ϕ is an infinitary propositional formula and $\kappa > \text{rank}(\phi)$ is inaccessible. Then ϕ is said to be AS-good at κ iff ϕ is satisfiable and for all stationary $S \subseteq \omega_1$, it holds in $V^{\text{Col}(\omega, < \kappa)}$ that for all $\mu \models \phi$, there exist elementary $\tau : V \longrightarrow W$ and $\hat{\mu} \models \tau(\phi)$ such that $\text{crit}(\tau) = \omega_1^V \in \tau(S)$ and for all subformulas ψ of ϕ , $\hat{\mu}(\tau(\psi)) = \mu(\psi)$.

Theorem (K., Veličković)

Suppose κ is an inaccessible cardinal and $\phi \in V_\kappa$ is an infinitary propositional formula which is AS-good at κ . Then there exists an ssp poset that forces a model for ϕ .

Thank you for your attention!